5.

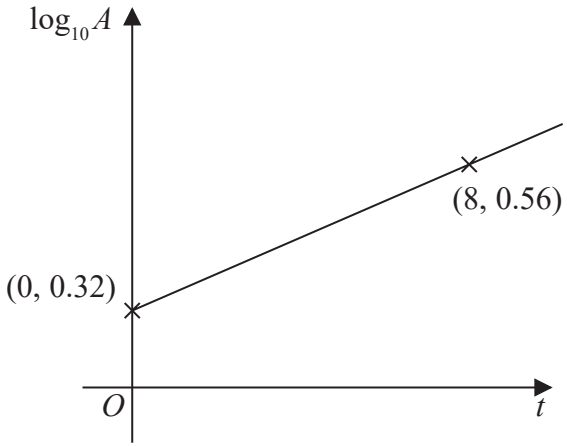


Figure 1

The growth of duckweed on a pond is being studied.

The surface area of the pond covered by duckweed, $A\text{m}^2$, at a time t days after the start of the study is modelled by the equation

$$A = pq^t \qquad \text{where } p \text{ and } q \text{ are positive constants}$$

Figure 1 shows the linear relationship between $\log_{10} A$ and t.

The points (0, 0.32) and (8, 0.56) lie on the line as shown.

(a) Find, to 3 decimal places, the value of p and the value of q.

(4)

Using the model with the values of p and q found in part (a),

(b) find the rate of increase of the surface area of the pond covered by duckweed, in m^2 / day, exactly 6 days after the start of the study.
Give your answer to 2 decimal places.

(3)